

UCS534: COMPUTER & NETWORK SECURITY

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Course Objectives: This course is designed to impart a critical theoretical and detailed practical knowledge of a range of computer network security technologies as well as network security tools.

Introduction: Security Attacks, Security Services, Security Mechanisms and Principles, Security goals, Malicious software, Worms, Viruses, Trojans, Spyware, Botnets, Life cycle of a vulnerability: CAN and CVE.

Computer Security: Set-UID programs, privileged programs, environment variables: hidden inputs, capability leaking, invoking other programs, principle of least privileges. Environment variables and attacks, attacks via dynamic linker, external program and library. Shellshock attack, exploiting shellshock vulnerability. Buffer overflow attacks: program memory layout, stack and function invocation. Writing a shell code, injecting code into buffer, address space layout randomization, Stack Guard.

Network Security: Packet sniffing and spoofing, Attacks on TCP protocol, SYN flood, TCP reset attack, session hijacking attack, Firewalls: Packet filter, Stateful firewall, Application firewall. IP tables, DNS poisoning, Authoritative replies, ARP poisoning, Heartbleed Bug and Attack, Public key infrastructure and Transport Layer Security.

Laboratory work:

Demonstrate use of Environment variables and privileged programs, Demonstrate Buffer Overflow and showcase EIP and other register status, insert malicious shell code into a program file and check its malicious or benign status, perform ARP poisoning, implement stateful firewall using IPTables.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Identify software vulnerabilities and apply various security mechanisms to protect against security attacks.
2. Demonstrate shellshock attack and its countermeasure.
3. Demonstrate buffer-overflow attack, locate and fix security leaks in a computer software.
4. Implement firewall and its variants.
5. Implement PKI and TLS.

Text Books:

1. Stallings, W., Network Security Essentials, Prentice Hall (2017) 6th Edition.
2. Cheswick, R., W., Bellovin, M., S., and Rubin, D., A., Firewalls and Internet Security, Addison-Wesley Professional (2003) 2nd Edition.
3. Wenliang Du, Computer Security: A hands-on approach, CreateSpace (2017).

Reference Books:

1. Graves, K., Certified Ethical Hacking Study Guide, Sybex (2010) 1st Edition.
2. Stallings, W., Cryptography and Network Security, Prentice Hall (2013), 6th Edition.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments//Quizzes)	30